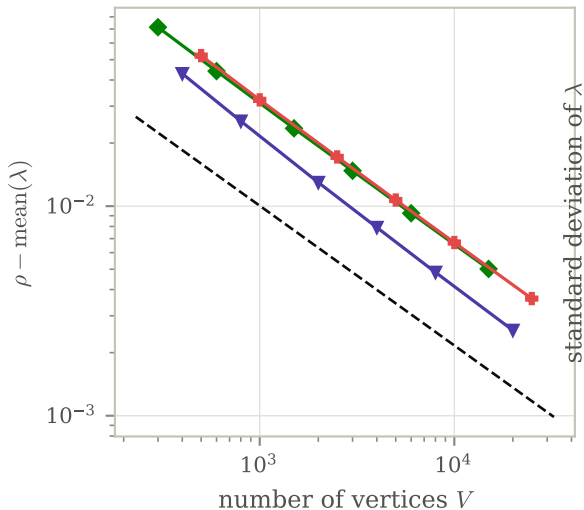


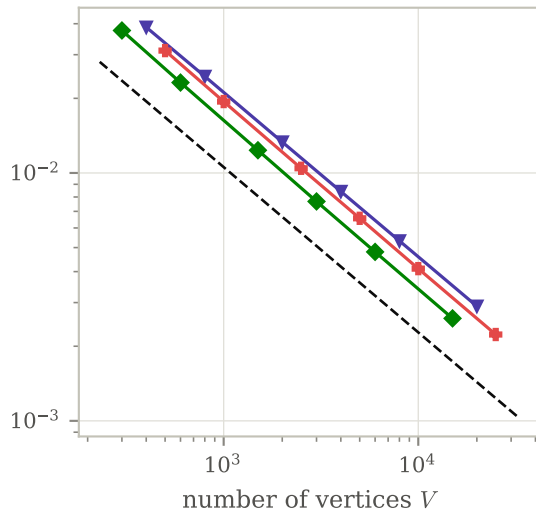
$V^{-2/3}$ scaling of the largest new eigenvalue — abelian covers ($V = \text{base} \times k$)

Gap of the mean below the Ramanujan threshold ρ

Standard deviation



- ◆ Abelian cover, $k=3$ (slope -0.679)
- ▼ Abelian cover, $k=4$ (slope -0.721)
- Abelian cover, $k=5$ (slope -0.682)
- $V^{-2/3}$ slope guide (regular-graph amplitude)



- ◆ Abelian cover, $k=3$ (slope -0.684)
- ▼ Abelian cover, $k=4$ (slope -0.663)
- Abelian cover, $k=5$ (slope -0.674)
- $V^{-2/3}$ slope guide (regular-graph amplitude)